

SIMULATION IDEAS

For each observation $i \in [n]$, denote by its tensor covariate $\mathcal{X}_i \in \mathbb{R}^{p_1 \times \dots \times p_D}$, one-way covariates $\mathbf{z}_i \in \mathbb{R}^{p_0}$, and two responses $y_{ij} \in \mathbb{R}$ where $j = 1, 2$ represents two eyes. Let $t \in [0, 1]$ be the observation time, $\mathcal{A}(t) = (\alpha_{j_1, \dots, j_d}(t)) \in \mathbb{R}^{p_1 \times \dots \times p_D}$ be the unknown tensor coefficient function, and $\boldsymbol{\beta} \in \mathbb{R}^{p_0}$ be the coefficients. The model is assumed to be

$$y_{ij} = \langle \mathcal{X}_{ij}, \mathcal{A}(t_i) \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} = \sum_{s=1}^S \left\langle \mathcal{X}_{ij}^{(s)}, \mathcal{A}^{(s)}(t_i) \right\rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} \quad (1)$$

by a rank- R CP decomposition, we can reorganize it in a matrix form

$$\approx \sum_{s=1}^S \sum_{r=1}^R x_{ijrs}^* \left\langle \mathcal{U}_r^{(s)}, \mathcal{A}^{(s)}(t_i) \right\rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} = \langle \mathbf{X}_{ij}^*, \mathbf{A}(t_i) \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij},$$

where matrix $\mathbf{X}_{ij}^* := (x_{ijrs}^*) \in \mathbb{R}^{R \times S}$, and $\mathbf{A}(t) := (\alpha_{rs}(t)) \in \mathbb{R}^{R \times S}$ with $\alpha_{rs}(t) := \left\langle \mathcal{U}_r^{(s)}, \mathcal{A}^{(s)}(t) \right\rangle$. Within one patient, the eyes are correlated by

$$\begin{bmatrix} \epsilon_{i1} \\ \epsilon_{i2} \end{bmatrix} \sim \mathbf{N} \left(\mathbf{0}, \begin{bmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{bmatrix} \right).$$

1 Paired Simulation for Case I

DGP. The tensor size is set to be $D = 2$, $p_1 = p_2 = 80$, and $S = 16$. We split the tensor by $S^{(1)} = S^{(2)} = 4$ and $p'_1 = p'_2 = 20$. Set the rank $R = 10$, and the s -th block tensor for j -th eye of i -th patient is generated by:

$$\mathcal{X}_{ij}^{(s)} = \sum_{r=1}^R x_{ijrs}^* \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)}, \text{ where } x_{ijrs}^* \stackrel{\text{iid}}{\sim} \mathbf{N}(0, 1) \quad \textbf{Remark. Can consider correlation between } j = 1, 2.$$

and any $\mathbf{x}_r^{(sd)}$ and $\mathbf{x}_{r'}^{(sd)}$ are orthonormal and obtained from the QR decomposition of a random matrix. The coefficient function is set to be $\mathcal{A}^{(s)}(t) = \sum_{r=1}^R \alpha_{rs}(t) \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)}$ where $\alpha_{rs}(t)$ has 4 choices:

$$\alpha_{rs}(t) = 4\sqrt{\frac{rs}{RS}} \times (t - 0.5)^2 \rightarrow \mathcal{A}_1(t) \text{ and } \mathcal{A}_2(t) \leftarrow \alpha_{rs}(t) = \sqrt{\frac{rs}{RS}} \times t^{0.5}, \quad (2)$$

$$\alpha_{rs}(t) = 1.75\sqrt{\frac{rs}{RS}} \left(e^{-(3t-1)^2} + e^{-(4t-3)^2} - 0.75 \right) \rightarrow \mathcal{A}_3(t) \text{ and } \mathcal{A}_4(t) \leftarrow \alpha_{rs}(t) = \sqrt{\frac{rs}{RS}} \sin(2\pi(t - 0.5)). \quad (3)$$

The one-way vector $\mathbf{Z} \in \mathbb{R}^{n \times p_0}$ is Gaussian ensemble matrix (iid entries) and $\boldsymbol{\beta} = (3, 3)^\top$. Finally the responses are generated by model eq(1).

Estimation Steps. (1) We treat the paired data as iid, and get initial estimate $\hat{\mathbf{A}}^\dagger(t)$ and $\hat{\boldsymbol{\beta}}^\dagger$. (2) Then estimate

$$\hat{\sigma}^2 = \frac{\sum_i \sum_j \left[y_{ij} - \left\langle \mathbf{X}_{ij}^*, \hat{\mathbf{A}}^\dagger(t_i) \right\rangle - \mathbf{z}_i^\top \hat{\boldsymbol{\beta}}^\dagger \right]^2}{2n}, \quad \hat{\rho} = \frac{\sum_{i=1}^n \left[y_{i1} - \left\langle \mathbf{X}_{i1}^*, \hat{\mathbf{A}}^\dagger(t_i) \right\rangle - \mathbf{z}_i^\top \hat{\boldsymbol{\beta}}^\dagger \right] \left[y_{i2} - \left\langle \mathbf{X}_{i2}^*, \hat{\mathbf{A}}^\dagger(t_i) \right\rangle - \mathbf{z}_i^\top \hat{\boldsymbol{\beta}}^\dagger \right]}{n\hat{\sigma}^2}.$$

(3) Get final estimates $\hat{\mathbf{A}}^*(t)$ and $\hat{\boldsymbol{\beta}}^*$ from Eq(14) and Eq(15) of AOAS draft.

Simulation Results. We vary $n = 500/1000/2000$, coefficients $\mathcal{A}_1(t), \dots, \mathcal{A}_4(t)$, and correlation $\rho = 0/0.3/0.6/0.9$. We report the error of both initial estimates $(\hat{\mathbf{A}}^\dagger, \hat{\boldsymbol{\beta}}^\dagger)$ and final estimates $(\hat{\mathbf{A}}^*, \hat{\boldsymbol{\beta}}^*)$, also the error of $(\hat{\sigma}^2, \hat{\rho})$.

Tables. All values in the following tables are reported in units of $\times 10^{-2}$; Under each correlation, the better value between phase 1/3 is highlighted.

Table 1: $\mathbf{A}(t)$ MIAE: phase 1 vs phase 3.

function	n	$\rho = 0$		$\rho = 0.3$		$\rho = 0.6$		$\rho = 0.9$	
		phase1	phase3	phase1	phase3	phase1	phase3	phase1	phase3
$\mathbf{A}_1(t)$	500	6.60 (± 0.46)	6.61 (± 0.46)	6.72 (± 0.45)	6.42 (± 0.47)	6.71 (± 0.43)	5.50 (± 0.39)	6.69 (± 0.40)	3.23 (± 0.22)
	1000	4.35 (± 0.26)	4.35 (± 0.26)	4.55 (± 0.33)	4.36 (± 0.31)	4.53 (± 0.33)	3.69 (± 0.26)	4.48 (± 0.34)	2.07 (± 0.14)
	2000	3.12 (± 0.20)	3.12 (± 0.20)	3.09 (± 0.20)	2.95 (± 0.20)	3.08 (± 0.20)	2.49 (± 0.18)	3.07 (± 0.20)	1.40 (± 0.10)
$\mathbf{A}_2(t)$	500	6.62 (± 0.44)	6.62 (± 0.45)	6.74 (± 0.45)	6.44 (± 0.46)	6.73 (± 0.43)	5.52 (± 0.39)	6.70 (± 0.41)	3.27 (± 0.22)
	1000	4.40 (± 0.28)	4.40 (± 0.28)	4.60 (± 0.36)	4.42 (± 0.35)	4.57 (± 0.36)	3.75 (± 0.30)	4.51 (± 0.36)	2.17 (± 0.18)
	2000	3.16 (± 0.21)	3.16 (± 0.21)	3.10 (± 0.20)	2.97 (± 0.21)	3.10 (± 0.20)	2.52 (± 0.19)	3.10 (± 0.20)	1.48 (± 0.12)
$\mathbf{A}_3(t)$	500	6.79 (± 0.51)	6.79 (± 0.51)	6.88 (± 0.46)	6.59 (± 0.47)	6.88 (± 0.45)	5.70 (± 0.39)	6.86 (± 0.44)	3.58 (± 0.22)
	1000	4.58 (± 0.29)	4.58 (± 0.29)	4.78 (± 0.32)	4.59 (± 0.27)	4.76 (± 0.34)	3.95 (± 0.22)	4.72 (± 0.35)	2.50 (± 0.12)
	2000	3.45 (± 0.21)	3.45 (± 0.21)	3.48 (± 0.20)	3.36 (± 0.20)	3.48 (± 0.21)	2.95 (± 0.18)	3.47 (± 0.21)	2.04 (± 0.11)
$\mathbf{A}_4(t)$	500	7.04 (± 0.55)	7.04 (± 0.56)	7.08 (± 0.53)	6.81 (± 0.52)	7.07 (± 0.51)	5.97 (± 0.44)	7.06 (± 0.45)	4.06 (± 0.27)
	1000	4.99 (± 0.41)	4.99 (± 0.41)	5.15 (± 0.40)	4.99 (± 0.37)	5.11 (± 0.41)	4.41 (± 0.31)	5.05 (± 0.40)	3.16 (± 0.21)
	2000	3.84 (± 0.28)	3.84 (± 0.28)	3.89 (± 0.20)	3.77 (± 0.21)	3.90 (± 0.19)	3.43 (± 0.20)	3.90 (± 0.21)	2.76 (± 0.13)

Table 2: β MAE: phase 1 vs phase 3.

function	n	$\rho = 0$		$\rho = 0.3$		$\rho = 0.6$		$\rho = 0.9$	
		phase1	phase3	phase1	phase3	phase1	phase3	phase1	phase3
$\mathbf{A}_1(t)$	500	2.85 (± 1.32)	2.84 (± 1.32)	3.00 (± 1.58)	2.92 (± 1.57)	3.30 (± 1.73)	3.13 (± 1.67)	3.57 (± 1.88)	3.26 (± 1.76)
	1000	2.07 (± 0.78)	2.07 (± 0.78)	2.03 (± 0.89)	2.06 (± 0.89)	2.23 (± 0.98)	2.28 (± 0.99)	2.40 (± 1.07)	2.47 (± 1.07)
	2000	1.43 (± 0.60)	1.43 (± 0.60)	1.83 (± 0.82)	1.82 (± 0.81)	2.03 (± 0.91)	2.01 (± 0.88)	2.20 (± 0.99)	2.16 (± 0.94)
$\mathbf{A}_2(t)$	500	2.84 (± 1.32)	2.83 (± 1.33)	3.01 (± 1.59)	2.93 (± 1.57)	3.31 (± 1.73)	3.14 (± 1.67)	3.58 (± 1.87)	3.27 (± 1.76)
	1000	2.06 (± 0.80)	2.06 (± 0.80)	2.02 (± 0.90)	2.05 (± 0.90)	2.21 (± 1.00)	2.27 (± 1.00)	2.39 (± 1.08)	2.46 (± 1.07)
	2000	1.43 (± 0.61)	1.44 (± 0.61)	1.84 (± 0.82)	1.83 (± 0.80)	2.03 (± 0.90)	2.01 (± 0.87)	2.20 (± 0.98)	2.17 (± 0.93)
$\mathbf{A}_3(t)$	500	2.83 (± 1.32)	2.81 (± 1.32)	3.03 (± 1.58)	2.95 (± 1.55)	3.33 (± 1.73)	3.15 (± 1.67)	3.60 (± 1.87)	3.28 (± 1.76)
	1000	2.10 (± 0.83)	2.10 (± 0.83)	2.00 (± 0.88)	2.03 (± 0.88)	2.20 (± 0.97)	2.25 (± 0.98)	2.36 (± 1.06)	2.43 (± 1.07)
	2000	1.45 (± 0.59)	1.45 (± 0.59)	1.84 (± 0.79)	1.83 (± 0.78)	2.03 (± 0.88)	2.01 (± 0.85)	2.20 (± 0.96)	2.16 (± 0.91)
$\mathbf{A}_4(t)$	500	2.83 (± 1.37)	2.82 (± 1.38)	3.01 (± 1.57)	2.93 (± 1.56)	3.31 (± 1.72)	3.13 (± 1.69)	3.58 (± 1.85)	3.26 (± 1.78)
	1000	2.02 (± 0.86)	2.02 (± 0.86)	2.02 (± 0.93)	2.04 (± 0.94)	2.22 (± 1.02)	2.26 (± 1.02)	2.39 (± 1.11)	2.46 (± 1.09)
	2000	1.48 (± 0.63)	1.49 (± 0.64)	1.82 (± 0.78)	1.81 (± 0.77)	2.01 (± 0.87)	1.99 (± 0.84)	2.18 (± 0.95)	2.14 (± 0.90)

- (i) When $\rho = 0$, there is no difference between iid estimation ($\hat{\mathbf{A}}^\dagger(t), \hat{\beta}^\dagger$) and paired estimation ($\hat{\mathbf{A}}^*(t), \hat{\beta}^*$).
- (ii) The larger the ρ is, the better the paired estimation $\hat{\mathbf{A}}^*(t)$ is than iid $\hat{\mathbf{A}}^\dagger(t)$.
- (iii) The larger the ρ is, the harder to estimate β for both paired estimation $\hat{\beta}^*$ and iid estimation $\hat{\beta}^\dagger$.
- (iv) All estimation errors decrease as n increases.

Table 3: $\mathbf{A}(t)$ RMISE: phase 1 vs phase 3.

function	n	$\rho = 0$		$\rho = 0.3$		$\rho = 0.6$		$\rho = 0.9$	
		phase1	phase3	phase1	phase3	phase1	phase3	phase1	phase3
$\mathbf{A}_1(t)$	500	8.70 (± 0.61)	8.71 (± 0.62)	8.83 (± 0.56)	8.49 (± 0.56)	8.81 (± 0.53)	7.32 (± 0.49)	8.78 (± 0.51)	4.39 (± 0.30)
	1000	5.68 (± 0.32)	5.68 (± 0.32)	5.99 (± 0.47)	5.74 (± 0.42)	5.95 (± 0.48)	4.87 (± 0.36)	5.88 (± 0.49)	2.75 (± 0.20)
	2000	4.09 (± 0.25)	4.09 (± 0.24)	4.02 (± 0.28)	3.85 (± 0.28)	4.01 (± 0.28)	3.26 (± 0.25)	4.01 (± 0.27)	1.84 (± 0.14)
$\mathbf{A}_2(t)$	500	8.72 (± 0.59)	8.73 (± 0.60)	8.84 (± 0.56)	8.50 (± 0.56)	8.81 (± 0.54)	7.33 (± 0.49)	8.78 (± 0.52)	4.42 (± 0.32)
	1000	5.74 (± 0.34)	5.74 (± 0.34)	6.02 (± 0.47)	5.78 (± 0.44)	5.98 (± 0.49)	4.91 (± 0.38)	5.91 (± 0.50)	2.82 (± 0.22)
	2000	4.13 (± 0.26)	4.13 (± 0.26)	4.04 (± 0.29)	3.86 (± 0.29)	4.03 (± 0.28)	3.28 (± 0.26)	4.04 (± 0.27)	1.91 (± 0.15)
$\mathbf{A}_3(t)$	500	8.89 (± 0.66)	8.89 (± 0.67)	9.00 (± 0.60)	8.67 (± 0.60)	8.98 (± 0.56)	7.53 (± 0.52)	8.95 (± 0.53)	4.75 (± 0.33)
	1000	5.93 (± 0.34)	5.93 (± 0.34)	6.23 (± 0.46)	5.99 (± 0.40)	6.19 (± 0.48)	5.16 (± 0.32)	6.12 (± 0.50)	3.25 (± 0.17)
	2000	4.45 (± 0.24)	4.45 (± 0.24)	4.45 (± 0.27)	4.30 (± 0.26)	4.44 (± 0.28)	3.78 (± 0.23)	4.43 (± 0.28)	2.62 (± 0.13)
$\mathbf{A}_4(t)$	500	9.15 (± 0.66)	9.16 (± 0.67)	9.22 (± 0.62)	8.90 (± 0.62)	9.20 (± 0.57)	7.82 (± 0.52)	9.19 (± 0.52)	5.25 (± 0.33)
	1000	6.37 (± 0.48)	6.37 (± 0.47)	6.60 (± 0.51)	6.38 (± 0.46)	6.56 (± 0.52)	5.60 (± 0.38)	6.49 (± 0.51)	3.93 (± 0.24)
	2000	4.87 (± 0.33)	4.87 (± 0.33)	4.90 (± 0.27)	4.74 (± 0.27)	4.90 (± 0.27)	4.28 (± 0.24)	4.91 (± 0.28)	3.35 (± 0.15)

Table 4: β RMSE: phase 1 vs phase 3.

function	n	$\rho = 0$		$\rho = 0.3$		$\rho = 0.6$		$\rho = 0.9$	
		phase1	phase3	phase1	phase3	phase1	phase3	phase1	phase3
$\mathbf{A}_1(t)$	500	3.18 (± 1.44)	3.16 (± 1.45)	3.25 (± 1.71)	3.19 (± 1.69)	3.58 (± 1.85)	3.42 (± 1.80)	3.87 (± 1.98)	3.58 (± 1.87)
	1000	2.27 (± 0.87)	2.27 (± 0.87)	2.39 (± 1.07)	2.41 (± 1.06)	2.63 (± 1.19)	2.67 (± 1.18)	2.83 (± 1.31)	2.89 (± 1.28)
	2000	1.60 (± 0.63)	1.60 (± 0.64)	2.07 (± 0.91)	2.06 (± 0.90)	2.29 (± 1.01)	2.27 (± 0.98)	2.49 (± 1.10)	2.45 (± 1.06)
$\mathbf{A}_2(t)$	500	3.17 (± 1.43)	3.15 (± 1.44)	3.26 (± 1.71)	3.20 (± 1.69)	3.58 (± 1.86)	3.43 (± 1.80)	3.87 (± 1.98)	3.59 (± 1.87)
	1000	2.27 (± 0.88)	2.27 (± 0.88)	2.37 (± 1.08)	2.39 (± 1.06)	2.61 (± 1.20)	2.65 (± 1.18)	2.82 (± 1.31)	2.88 (± 1.27)
	2000	1.60 (± 0.64)	1.60 (± 0.64)	2.07 (± 0.91)	2.06 (± 0.90)	2.30 (± 1.01)	2.27 (± 0.99)	2.50 (± 1.10)	2.46 (± 1.06)
$\mathbf{A}_3(t)$	500	3.14 (± 1.46)	3.13 (± 1.47)	3.26 (± 1.71)	3.19 (± 1.69)	3.58 (± 1.85)	3.42 (± 1.80)	3.88 (± 1.98)	3.58 (± 1.87)
	1000	2.31 (± 0.93)	2.31 (± 0.93)	2.35 (± 1.07)	2.38 (± 1.06)	2.59 (± 1.19)	2.63 (± 1.17)	2.79 (± 1.30)	2.85 (± 1.28)
	2000	1.61 (± 0.63)	1.61 (± 0.63)	2.06 (± 0.88)	2.05 (± 0.87)	2.29 (± 0.98)	2.26 (± 0.95)	2.49 (± 1.07)	2.45 (± 1.03)
$\mathbf{A}_4(t)$	500	3.15 (± 1.49)	3.13 (± 1.50)	3.25 (± 1.72)	3.19 (± 1.71)	3.56 (± 1.86)	3.41 (± 1.82)	3.85 (± 1.99)	3.57 (± 1.90)
	1000	2.25 (± 0.94)	2.24 (± 0.94)	2.37 (± 1.09)	2.39 (± 1.07)	2.60 (± 1.21)	2.64 (± 1.18)	2.81 (± 1.33)	2.87 (± 1.28)
	2000	1.64 (± 0.68)	1.65 (± 0.68)	2.04 (± 0.89)	2.03 (± 0.88)	2.26 (± 0.98)	2.24 (± 0.96)	2.46 (± 1.07)	2.43 (± 1.03)

Tables. All values in the following tables are reported in units of $\times 10^{-2}$; Similar results as MIAE for $\mathbf{A}(t)$ and MAE for β .

Table 5: Covariance estimation errors.

function	n	$ \hat{\sigma}^2 - \sigma^2 $				$ \hat{\rho} - \rho $			
		$\rho = 0$	$\rho = 0.3$	$\rho = 0.6$	$\rho = 0.9$	$\rho = 0$	$\rho = 0.3$	$\rho = 0.6$	$\rho = 0.9$
$\mathbf{A}_1(t)$	500	13.36 (± 3.61)	13.37 (± 4.37)	13.22 (± 4.90)	13.08 (± 5.49)	3.25 (± 2.36)	3.33 (± 2.62)	5.39 (± 2.73)	8.44 (± 1.49)
	1000	6.00 (± 3.43)	6.16 (± 3.35)	6.11 (± 3.67)	6.22 (± 4.09)	1.67 (± 1.44)	3.38 (± 2.24)	3.08 (± 2.20)	4.15 (± 0.97)
	2000	2.94 (± 1.77)	3.11 (± 1.89)	3.32 (± 2.27)	3.62 (± 2.62)	2.34 (± 1.56)	2.02 (± 1.59)	2.02 (± 1.44)	2.38 (± 0.54)
$\mathbf{A}_2(t)$	500	13.31 (± 3.60)	13.32 (± 4.38)	13.17 (± 4.91)	13.04 (± 5.48)	3.31 (± 2.33)	3.35 (± 2.60)	5.45 (± 2.70)	8.51 (± 1.48)
	1000	5.93 (± 3.43)	6.09 (± 3.34)	6.03 (± 3.67)	6.16 (± 4.09)	1.65 (± 1.45)	3.40 (± 2.22)	3.07 (± 2.26)	4.20 (± 0.99)
	2000	2.88 (± 1.76)	3.04 (± 1.89)	3.25 (± 2.27)	3.57 (± 2.60)	2.34 (± 1.55)	2.04 (± 1.60)	2.07 (± 1.43)	2.42 (± 0.54)
$\mathbf{A}_3(t)$	500	12.93 (± 3.66)	12.93 (± 4.42)	12.79 (± 4.94)	12.68 (± 5.46)	3.21 (± 2.42)	3.40 (± 2.76)	5.65 (± 2.82)	8.80 (± 1.60)
	1000	5.61 (± 3.33)	5.74 (± 3.28)	5.70 (± 3.60)	5.89 (± 3.95)	1.67 (± 1.48)	3.45 (± 2.25)	3.29 (± 2.28)	4.58 (± 1.02)
	2000	2.51 (± 1.69)	2.68 (± 1.85)	2.97 (± 2.14)	3.34 (± 2.42)	2.31 (± 1.62)	2.03 (± 1.63)	2.24 (± 1.50)	2.84 (± 0.54)
$\mathbf{A}_4(t)$	500	12.36 (± 3.67)	12.34 (± 4.43)	12.19 (± 4.95)	12.13 (± 5.36)	3.41 (± 2.36)	3.63 (± 2.66)	5.98 (± 2.86)	9.32 (± 1.67)
	1000	5.09 (± 3.16)	5.17 (± 3.10)	5.14 (± 3.42)	5.41 (± 3.75)	1.71 (± 1.56)	3.44 (± 2.23)	3.52 (± 2.34)	5.15 (± 0.96)
	2000	1.98 (± 1.49)	2.12 (± 1.69)	2.50 (± 1.89)	2.89 (± 2.19)	2.37 (± 1.65)	2.14 (± 1.76)	2.74 (± 1.49)	3.52 (± 0.56)

For covariance estimation, both $\hat{\sigma}^2$ and $\hat{\rho}$ are estimated with reasonable accuracy. And

- The larger the ρ is, the harder to estimate it.
- The larger the n is, the smaller the error is.

2 Paired Simulation for Case II

DGP. The tensor size is set to be $D = 3$, $p_1 = p_2 = p_3 = 40$, and $S = 64$. We split the tensor by $S^{(1)} = S^{(2)} = S^{(3)} = 10$ and $p'_1 = p'_2 = p'_3 = 4$. Set the rank $R = 5$, and covariate tensor:

$$\mathcal{X}_{ij}^{(s)} = \sum_{r=1}^R x_{ijrs}^* \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)} \circ \mathbf{x}_r^{(s3)}, \text{ where } x_{ijrs}^* \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$$

and any $\mathbf{x}_r^{(sd)}$ and $\mathbf{x}_{r'}^{(sd)}$ are orthonormal and obtained from the QR decomposition of a random matrix. The coefficient function is set to be $\mathcal{A}^{(s)}(t) = \sum_{r=1}^R \alpha_{rs}(t) \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)} \circ \mathbf{x}_r^{(s3)}$ where $\alpha_{rs}(t)$ has 4 choices in Eq(2) and (3). Everything else is same as Case I.

Simulation Results. We vary $n = 500/1000/2000$, coefficients $\mathcal{A}_1(t), \dots, \mathcal{A}_4(t)$, and correlation $\rho = 0/0.3/0.6/0.9$. We report the error of both initial estimates ($\hat{\mathbf{A}}^\dagger, \hat{\beta}^\dagger$) and final estimates ($\hat{\mathbf{A}}^*, \hat{\beta}^*$), also the error of $(\hat{\sigma}^2, \hat{\rho})$.

Simulations to be done...